**This assignment will not be graded and is only for practice.**

**Level 1**

1) Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times 3}(\mathrm{R})$ ) in column A with their bases in column B in Table : M2W4AQ2. ***1 point***

| | Vector space (Column A) | | Basis (Column B) |
|---|---|---|---|
| a) | $V = \left\{ \begin{bmatrix} x & y & 0 \\ 0 & y & 0 \\ 0 & x & x \end{bmatrix} \mid x+y=0, \text{ and } x, y \in \mathbb{R} \right\}$ | i) | $\left\{ \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right\}$ |
| b) | $V = \{A \mid A \in M_{3\times 3}(\mathbb{R}),\ A \text{ is a diagonal matrix, and the sum of the diagonal entries is zero. }\}$ | ii) | $\left\{ \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix} \right\}$ |
| c) | $V = \left\{ \begin{bmatrix} 0 & y & x \\ 0 & 0 & y \\ 0 & 0 & 0 \end{bmatrix} \mid x+y=0, \text{ and } x, y \in \mathbb{R} \right\}$ | iii) | $\left\{ \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\}$ |

Table : M2W4AQ2

Choose the correct option.

- [ ] a → ii
- [ ] a → i
- [ ] b → iii
- [ ] c → i
- [ ] b → ii
- [ ] c → iii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a → ii
b → iii
c → i

2) If $S$ be a subset of $\mathrm{R}^5$ such that $span(S) = \mathrm{R}^5$, then what is the minimum number of elements possible in $S$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

***1 point***

3) If $S_1$ is a maximal linear independent set and $S_2$ is a minimal spanning set of a vector space $V$, then which of the following option(s) is (are) true? ***1 point***

- [ ] For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
- [ ] For any $v \in S_2, S_2 \setminus \{v\}$ is a spanning set of $V$.
- [ ] For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
- [ ] For any $v \in V, S_2 \cup \{v\}$ is a spanning set of $V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
For any $v \in V, S_2 \cup \{v\}$ is a spanning set of $V$.

4) Consider the following subset of $\mathrm{R}^2$ with usual addition and scalar multiplication as in $\mathrm{R}^2$. ***1 point***

$V_1 = \{(x, 0) \mid x \in \mathrm{R}\}$
$V_2 = \{(0, y) \mid y \in \mathrm{R}\}$
Which of the following options are correct?

- [ ] Both $V_1$ and $V_2$ are subspaces of $\mathrm{R}^2$.
- [ ] $V_1 \cap V_2$ is a subspace of $\mathrm{R}^2$.
- [ ] $V_1 \cup V_2$ is a subspace of $\mathrm{R}^2$.
- [ ] $V_1$ is a subspace of $\mathrm{R}^2$, but $V_2$ is not.
- [ ] $V_2$ is a subspace of $\mathrm{R}^2$, but $V_1$ is not.
- [ ] $\{(1, 0)\}$ is a basis of $V_1$.
- [ ] $\{(0, 1)\}$ is a basis of $V_2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both $V_1$ and $V_2$ are subspaces of $\mathrm{R}^2$.
$V_1 \cap V_2$ is a subspace of $\mathrm{R}^2$.
$\{(1, 0)\}$ is a basis of $V_1$.
$\{(0, 1)\}$ is a basis of $V_2$.

5) Consider the following subset of $\mathrm{R}^2$ with usual addition and scalar multiplication as in $\mathrm{R}^2$. ***1 point***
$V_1 = \{(x, 0) \mid x \in \mathrm{R}\}$
$V_2 = \{(2x, 0) \mid x \in \mathrm{R}\}$
Which of the following options are correct?

- [ ] Both $V_1$ and $V_2$ are subspaces of $\mathrm{R}^2$.
- [ ] $V_1 \cap V_2$ is a subspace of $\mathrm{R}^2$.
- [ ] $V_1 \cup V_2$ is a subspace of $\mathrm{R}^2$.
- [ ] $V_1$ is a subspace of $\mathrm{R}^2$, but $V_2$ is not.
- [ ] $V_2$ is a subspace of $\mathrm{R}^2$, but $V_1$ is not.
- [ ] $\{(1, 0)\}$ is a basis of $V_1$.

- [ ] $\{(1,0)\}$ is a basis of $V_2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both $V_1$ and $V_2$ are subspaces of $\mathrm{R}^2$.
$V_1 \cap V_2$ is a subspace of $\mathrm{R}^2$.
$V_1 \cup V_2$ is a subspace of $\mathrm{R}^2$.
$\{(1,0)\}$ is a basis of $V_1$.
$\{(1,0)\}$ is a basis of $V_2$.

6) Let $V$ be a vector space which is defined as follows: ***1 point***

$$V = \{(x,y,z,w) \mid x+z = y+w\} \subseteq \mathrm{R}^4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of $V$?

- [ ] $\{(1,0,0,0),(0,1,0,0),(0,0,0,1)\}$.
- [ ] $\{(1,1,0,0),(0,1,-1,0),(0,-1,0,1)\}$.
- [ ] $\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$.
- [ ] $\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$.
$\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

**Level 2**

7) Which of the following sets form a basis of the vector space of $2 \times 2$ lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct) ***1 point***

- [ ]

$\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\}$

- [ ] $\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\}$
- [ ] $\{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\}$
- [ ] $\{\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\}$
$\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\}$
$\{\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\}$

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8) If $S$ is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x, \text{ where } x, y \in \mathrm{R}\}$ with usual addition and scalar multiplication as in $\mathrm{R}^2$, then find the maximum possible cardinality of $S$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

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9) If $\{v_1, v_2, v_3\}$ forms a basis of $\mathrm{R}^3$, then which of the following are true? ***1 point***

- [ ] $\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathrm{R}^3$
- [ ] $\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathrm{R}^3$.
- [ ] $\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathrm{R}^3$.
- [ ] $\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathrm{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathrm{R}^3$
$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathrm{R}^3$.
$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathrm{R}^3$.
$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathrm{R}^3$.

10) Which of the following options is(are) true? ***1 point***

- [ ] Any minimal spanning set of a vector space $V$ must be a basis of $V$.
- [ ] Any maximal spanning set of a vector space $V$ must be a basis of $V$.
- [ ] Any minimal linear independent set of vector space $V$ must be a basis of $V$.
- [ ] Any maximal linear independent set of vector space $V$ must be a basis of $V$.
- [ ] The basis of a vector space is unique.
- [ ] The number of elements in a basis of $\mathrm{R}^3$ is $3$.
- [ ] The number of elements in a basis of $M_{3\times 3}(\mathrm{R})$ is $3$.
- [ ] There are infinite number of bases of $\mathrm{R}^3$.
- [ ] Any subset of a minimal spanning set of $V$ cannot be a spanning set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Any minimal spanning set of a vector space $V$ must be a basis of $V$.
Any maximal linear independent set of vector space $V$ must be a basis of $V$.
The number of elements in a basis of $\mathrm{R}^3$ is $3$.
There are infinite number of bases of $\mathrm{R}^3$.
Any subset of a minimal spanning set of $V$ cannot be a spanning set.

**Your score is: 0/10**